Protein Sequence Analysis and Functional Characterization Using Bioinformatics Tools

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Course:

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Abstract

This bioinformatics project analyzed a protein sequence of 95 amino acids to study its similarity, structure, and function among closely related species. **NCBI BLASTp** was used to find homologous sequences, **Clustal Omega** for multiple sequence alignment, and **Phylogeny.fr** to build a phylogenetic tree. The 3D structure was obtained from **RCSB PDB / AlphaFold** and visualized with **PyMOL** to examine folding and calcium-binding regions. Functional domains and physicochemical properties were identified using **InterPro** and **ProtParam**. **This study shows how bioinformatics tools can be used effectively to understand protein sequence, structure, and function.**

1. Introduction & Overview

Proteins are essential biomolecules that perform most of the biological functions in living organisms. Studying protein sequences helps us understand their structure, function, and importance in different species.

In this project, the given protein sequence:

MTELETAMGMIIDVFSRYSGSEGSTQTLTKGELKVLMEKELPGFLQSGKDKDAVDKL LKDLDANGDAQVDFSEFIVFVAAITSACHKYFEKAGLK

was analyzed using various **bioinformatics tools** to identify its homologs, study evolutionary relationships, and predict its structure and function.

The main objectives of this project are:

- To find homologous proteins among primates.
- To perform multiple sequence alignment (MSA) to identify conserved regions.

- To build a phylogenetic tree showing evolutionary relationships.
- To find and visualize the 3D structure of the protein.
- To determine the function and biological importance of the protein.

This project helps to apply bioinformatics concepts such as sequence analysis, structural prediction, and functional annotation, which are part of the **Principles of Bioinformatics (BTEN3051)** course.

2. Relevance to the Course

This project is directly related to the course **Principles of Bioinformatics (BTEN3051)**, as it involves the use of online bioinformatics tools to study proteins. It helps to understand how biological data, such as protein sequences, can be analyzed using computational methods.

By using tools like **BLASTp**, **Clustal Omega**, **Phylogeny.fr**, **RCSB PDB**, and **InterPro**, this project connects theoretical knowledge from the course with practical applications. Through this analysis, it becomes clear how bioinformatics helps in studying **sequence similarity, structural prediction, and functional annotation**.

Overall, this project strengthens understanding of how **bioinformatics integrates biology and computer science** to explore molecular and evolutionary relationships in living organisms.

3. Research Plan / Workflow

The following step-by-step plan was followed to analyze the given protein sequence using freely available online bioinformatics tools. Each step focused on a specific objective to understand the sequence, structure, and function of the protein.

Step 1: Homolog Search (BLASTp)

Objective: To find similar protein sequences among primates.
Tool Used: NCBI BLASTp
Method:

- The given protein sequence was entered in the BLASTp search box.
- The database selected was **nr (non-redundant)**, and the organism filter was set to **Primates [9443]**.
- The top homologous sequences were recorded based on **% identity** and **E-value**.

Step 2: Multiple Sequence Alignment (MSA)

Objective: To align homologous sequences and identify conserved regions.
Tool Used: Clustal Omega (Online)
Method:

- FASTA sequences of top homologs obtained from BLASTp were used as input.
- The alignment was performed using default parameters.
- Conserved residues and motifs were identified and highlighted.

Step 3: Phylogenetic Analysis

Objective: To study evolutionary relationships among primate sequences.
Tools Used: Phylogeny.fr / Phylo.io / NCBI BLAST Tree View
Method:

- The MSA file obtained from Clustal Omega was uploaded.
- A phylogenetic tree was generated using **Neighbor-Joining** and **Fast Minimum Evolution** methods.
- The tree was interpreted to identify closely related species.

Step 4: 3D Structure Prediction and Visualization

Objective: To obtain and visualize the 3D structure of the protein.
Tools Used: RCSB PDB, AlphaFold DB, PyMOL
Method:

- The protein was searched in RCSB PDB for an experimental structure.
- If unavailable, the predicted model from AlphaFold was used.
- The 3D model was viewed and analyzed using PyMOL.

Step 5: Functional and Biological Significance

Objective : To determine the protein's function and role in biological systems.
Tools Used: UniProt, InterPro, ProtParam
Method:

- The sequence was analyzed to find domains, motifs, and functions.
- Physicochemical properties such as molecular weight and pI were calculated.
- The protein's biological significance was interpreted based on results.

4.1 Homolog Search (BLASTp)

The given protein sequence was searched using the **NCBI BLASTp** tool to find similar sequences among **primates**.
The sequence was entered in FASTA format, and the search was carried out against the **nr (non-redundant)** protein database.
The **organism filter** was set to *Primates (TaxID: 9443)* to limit results to primate species.
The tool generated a list of homologous sequences with high similarity scores and very low E-values. The top eight sequences were selected for further analysis and alignment.

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) Clear

MTELETAMGMIIDVFSRYSGSEGSTQTLTKGELKVLMEKELPGFLQSGKDKDAV
DKLLKDLDANGDAQVDFSEFIVFVAAITSACHKYFEKAGLK

Query subrange

From

To

Or, upload file Choose File no file selected

Job Title

Enter a descriptive title for your BLAST search

Align two or more sequences

Choose Search Set

Database Non-redundant protein sequences (nr)

Organism Optional Primates (taxid:9443) exclude Add organism

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown.

Exclude Optional Models (XM/XP) Non-redundant RefSeq proteins (WP) Uncultured/environmental sample sequences

Program Selection

Algorithm
- Quick BLASTP (Accelerated protein-protein BLAST)
- blastp (protein-protein BLAST)
- PSI-BLAST (Position-Specific Iterated BLAST)
- PHI-BLAST (Pattern Hit Initiated BLAST)
- DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm

BLAST Search database nr using Blastp (protein-protein BLAST)

Show results in a new window

Your search is limited to records **include: Primates (taxid:9443)**

Job Title	Protein Sequence
RID	EWEWVBRH014 *Search expires on 10-15 20:01 pm* Download All
Program	BLASTP Citation
Database	nr See details
Query ID	lcl|Query_882965
Description	unnamed protein product
Molecule type	amino acid
Query Length	95
Other reports	Distance tree of results Multiple alignment MSA viewer

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to E value to Query Coverage to

Filter Reset

select all *8 sequences selected* GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
protein S100-P [Homo sapiens]	Homo sapiens	192	192	100%	1e-63	100.00%	95	NP_005971.1
protein S100-P [Papio anubis]	Papio anubis	192	192	100%	2e-63	100.00%	117	XP_009204751.1
PREDICTED: protein S100-P [Mandrillus leucophaeus]	Mandrillus leucophaeus	192	192	100%	2e-63	100.00%	117	XP_011857544.1
protein S100-P [Macaca mulatta]	Macaca mulatta	192	192	100%	2e-63	100.00%	117	XP_001091876.2
PREDICTED: protein S100-P [Colobus angolensis palliatus]	Colobus angolensis palliatus	192	192	100%	2e-63	100.00%	117	XP_011790797.1
PREDICTED: protein S100-P [Rhinopithecus bieti]	Rhinopithecus bieti	192	192	100%	2e-63	100.00%	117	XP_017708876.1
protein S100-P [Rhinopithecus roxellana]	Rhinopithecus roxellana	191	191	100%	6e-63	98.95%	117	XP_010368341.1
protein S100-P [Pan troglodytes]	Pan troglodytes	190	190	100%	7e-63	98.95%	95	XP_003310290.1

Screenshot of BLASTp results showing homologous protein sequences among primates.

Table 1: Top Homologous Sequences Identified by BLASTp

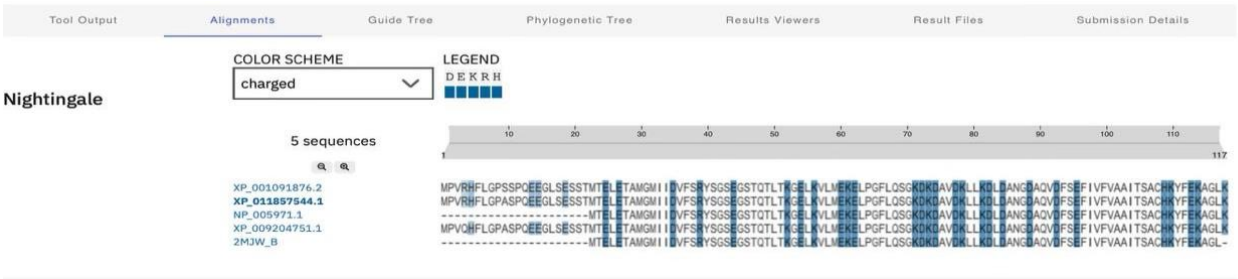
S.No	Species Name	Accession Number	% Identity	E-value	Length (aa)
1	*Homo sapiens* (Human)	NP_005971.1	100%	1E-63	95
2	*Papio anubis* (Olive baboon)	XP_009204751.1	100%	2E-63	117
3	*Mandrillus leucophaeus*	XP_011857544.1	100%	2E-63	117
4	*Macaca mulatta* (Rhesus macaque)	XP_001091876.2	100%	2E-63	117
5	*Colobus angolensis palliatus*	XP_011790797.1	100%	2E-63	117
6	*Rhinopithecus bieti*	XP_017708876.1	100%	2E-63	117
7	*Rhinopithecus roxellana*	XP_010368341.1	98.95%	6E-63	117
8	*Pan troglodytes* (Chimpanzee)	XP_003312090.1	98.95%	7E-63	95

4.2 Multiple Sequence Alignment (Clustal Omega / MEGA11)

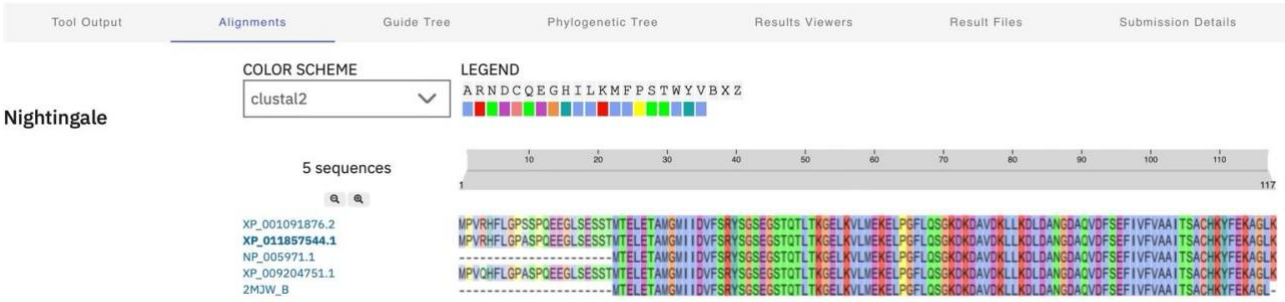
The protein sequences obtained from BLASTp were aligned using the **Clustal Omega** online tool.
The sequences were entered in **FASTA format** and aligned using **default settings**.
The alignment file was then downloaded and opened in **MEGA11 Alignment Explorer** for visualization.
Different color schemes such as **Clustal, Aromatic, Helix, and Charged** were used to highlight amino acid types.
The **Aromatic view** showed residues related to aromatic side chains.
The **Helix view** displayed the regions forming possible alpha-helices.
The **Charged view** showed positively and negatively charged amino acids clearly.
These color modes were used to study the arrangement and composition of residues in the aligned sequences.

```
CLUSTAL O(1.2.4) multiple sequence alignment


XP_001091876.2      MPVRHFLGPSSPQEEGLSESSTMTELETAMGMIIDVFSRYSGSEGSTQTLTKGELKVLME
XP_011857544.1      MPVRHFLGPASPQEEGLSESSTMTELETAMGMIIDVFSRYSGSEGSTQTLTKGELKVLME
NP_005971.1         ----------------------MTELETAMGMIIDVFSRYSGSEGSTQTLTKGELKVLME
XP_009204751.1      MPVQHFLGPASPQEEGLSESSTMTELETAMGMIIDVFSRYSGSEGSTQTLTKGELKVLME
2MJW_B              ----------------------MTELETAMGMIIDVFSRYSGSEGSTQTLTKGELKVLME
                                          **************************************

XP_001091876.2      KELPGFLQSGKDKDAVDKLLKDLDANGDAQVDFSEFIVFVAAITSACHKYFEKAGLK
XP_011857544.1      KELPGFLQSGKDKDAVDKLLKDLDANGDAQVDFSEFIVFVAAITSACHKYFEKAGLK
NP_005971.1         KELPGFLQSGKDKDAVDKLLKDLDANGDAQVDFSEFIVFVAAITSACHKYFEKAGLK
XP_009204751.1      KELPGFLQSGKDKDAVDKLLKDLDANGDAQVDFSEFIVFVAAITSACHKYFEKAGLK
2MJW_B              KELPGFLQSGKDKDAVDKLLKDLDANGDAQVDFSEFIVFVAAITSACHKYFEKAGL-
                    *******************************************************
```

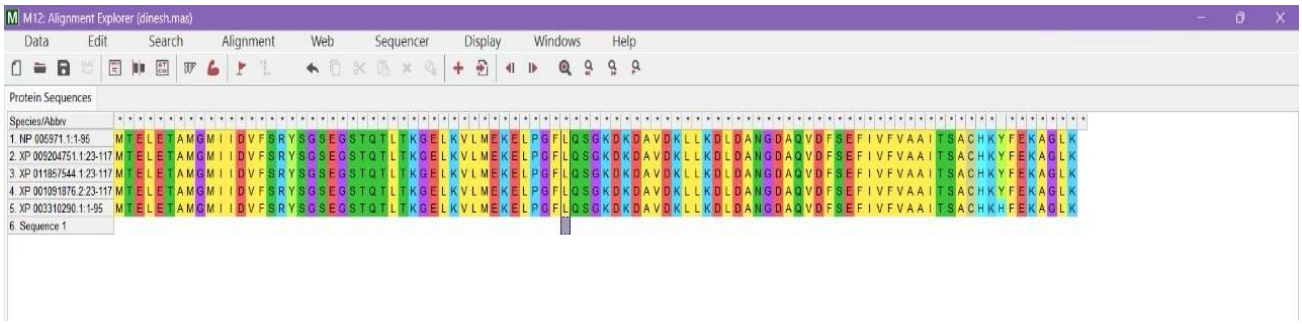
Multiple Sequence Alignment of primate protein sequences using Clustal Omega



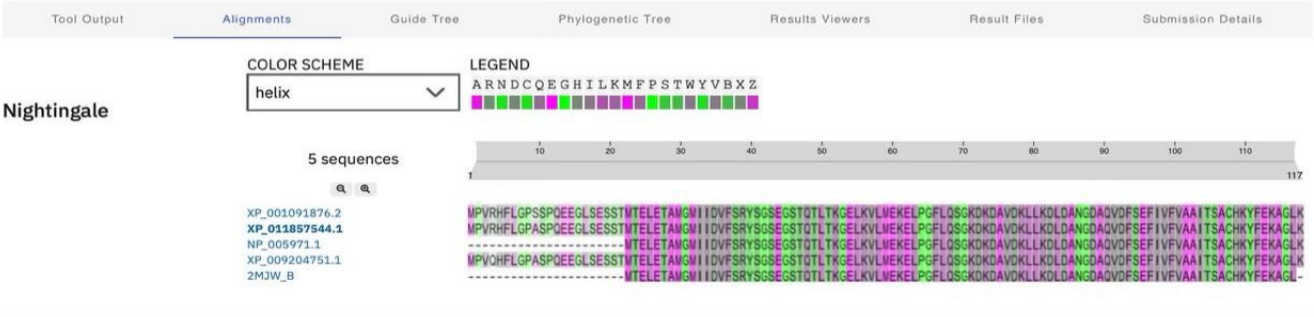
Clustal Omega alignment showing **positively and negatively charged residues** highlighted using the **Charged color scheme**.



Clustal Omega **default Clustal color scheme** view showing overall residue conservation among primate sequences.



Alignment visualization in MEGA11 showing conserved amino acid regions.



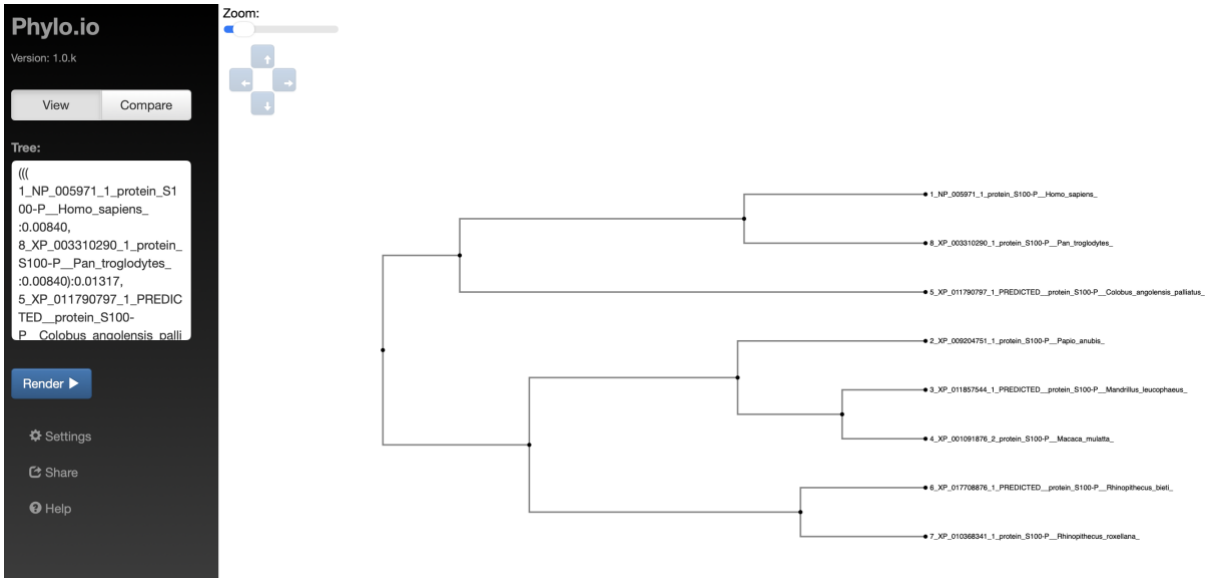
Helix color scheme view from Clustal Omega showing the predicted alpha-helical regions in the aligned protein sequences.

```
CLUSTAL O(1.2.4) multiple sequence alignment


XP_001091876.2      MPVRHFLGPSSPQEEGLSESSTMTELETAMGMIIDVFSRYSGSEGSTQTLTKGELKVLME
XP_011857544.1      MPVRHFLGPASPQEEGLSESSTMTELETAMGMIIDVFSRYSGSEGSTQTLTKGELKVLME
NP_005971.1         ----------------------MTELETAMGMIIDVFSRYSGSEGSTQTLTKGELKVLME
XP_009204751.1      MPVQHFLGPASPQEEGLSESSTMTELETAMGMIIDVFSRYSGSEGSTQTLTKGELKVLME
2MJW_B              ----------------------MTELETAMGMIIDVFSRYSGSEGSTQTLTKGELKVLME
                                          ***************************************

XP_001091876.2      KELPGFLQSGKDKDAVDKLLKDLDANGDAQVDFSEFIVFVAAITSACHKYFEKAGLK
XP_011857544.1      KELPGFLQSGKDKDAVDKLLKDLDANGDAQVDFSEFIVFVAAITSACHKYFEKAGLK
NP_005971.1         KELPGFLQSGKDKDAVDKLLKDLDANGDAQVDFSEFIVFVAAITSACHKYFEKAGLK
XP_009204751.1      KELPGFLQSGKDKDAVDKLLKDLDANGDAQVDFSEFIVFVAAITSACHKYFEKAGLK
2MJW_B              KELPGFLQSGKDKDAVDKLLKDLDANGDAQVDFSEFIVFVAAITSACHKYFEKAGL-
                    *******************************************************
```

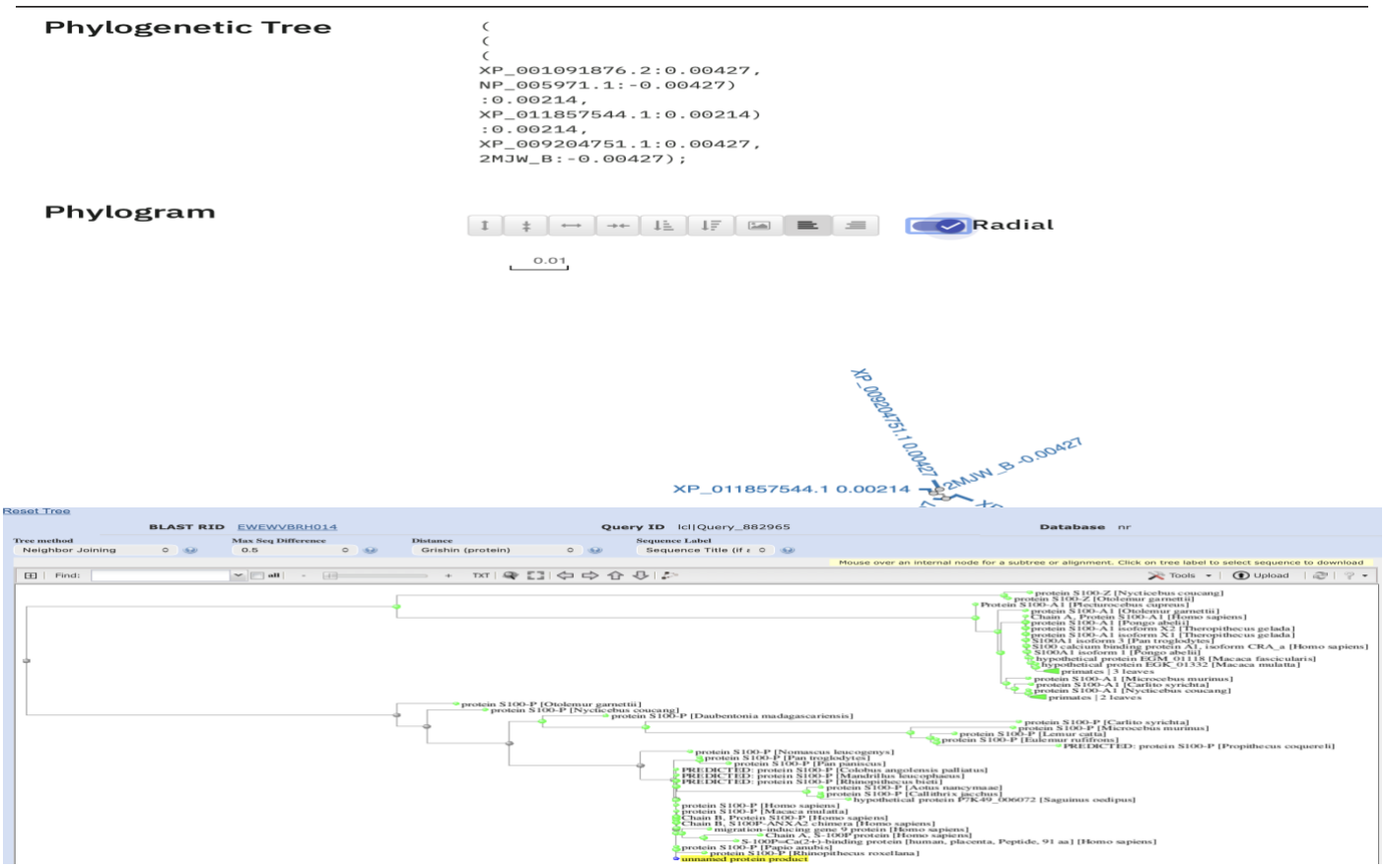
Multiple Sequence Alignment of primate protein sequences using Clustal Omega

4.3 Phylogenetic Analysis (Phylogeny.fr / Phylo.io / BLAST Tree View)

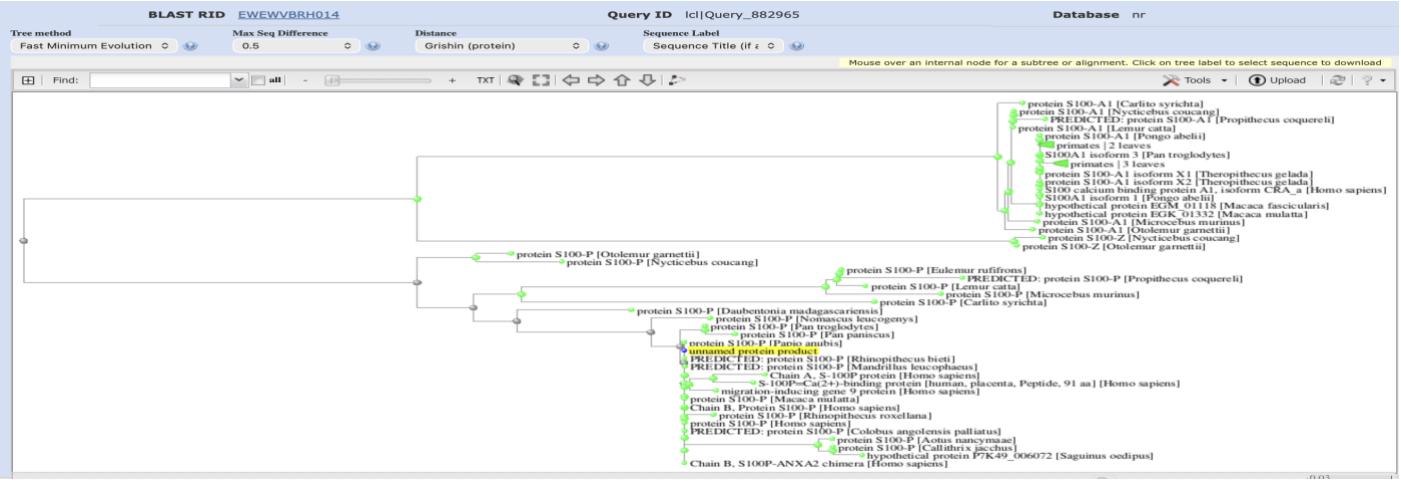
- The aligned protein sequences from **Clustal Omega** were used for **phylogenetic analysis**.
- The analysis was carried out using **Phylogeny.fr**, **Phylo.io**, and the **NCBI BLAST Tree View** tools.
- The uploaded **FASTA alignment** generated guide trees, phylograms, and evolutionary trees.
- The **Neighbor-Joining (NJ)** and **Fast Minimum Evolution (FME)** methods were used to create the BLAST tree.
- The generated trees grouped the sequences based on **evolutionary closeness and sequence similarity**.
- The trees were visualized using **Phylo.io** to clearly display the branching pattern among the primate protein sequences.



Phylogenetic tree viewed using **Phylo.io**, displaying the branching pattern and closeness of primate sequences.



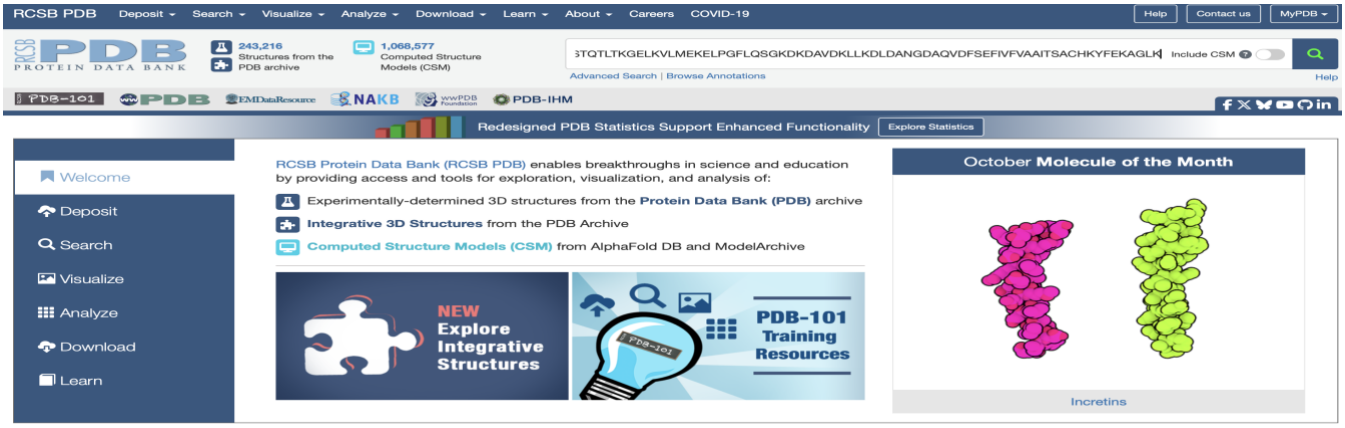
Phylogenetic tree generated using the **Neighbor-Joining (NJ)** method in **NCBI BLAST Tree View**, showing relationships among primate protein sequences.



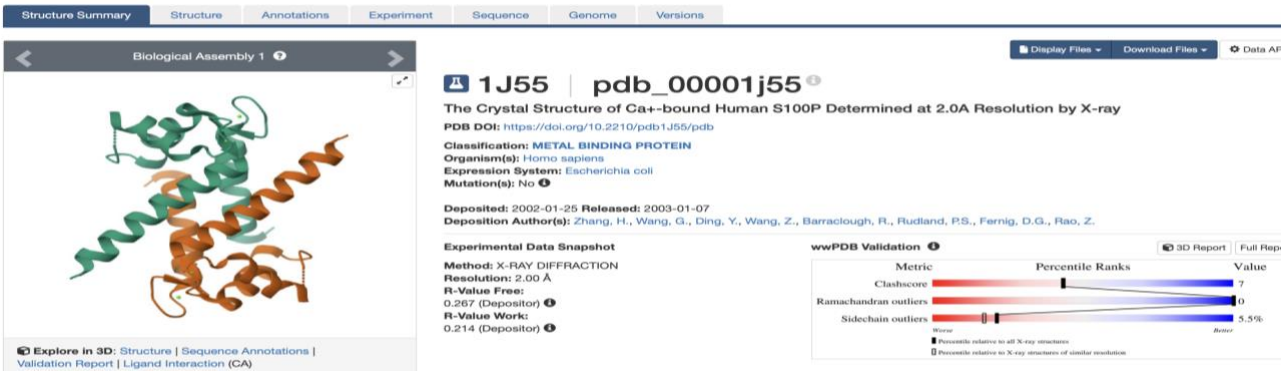
Phylogenetic tree (phylogram) generated using the Fast Minimum Evolution method in NCBI BLAST Tree View, showing relationships among primate proteins.

4.4 3D Structure Prediction (RCSB PDB)

- The 3D structure of the given protein was searched using the **RCSB Protein Data Bank (PDB)**.
- The sequence was entered in the PDB search bar, and matching structures were found.
- The protein with **PDB ID: 1J55** was identified as the **human S100P calcium-binding protein**.
- This structure was determined by the **X-ray diffraction method** at a **2.0 Å resolution**.
- The protein showed a **metal-binding property** and belonged to *Homo sapiens*.
- The model mainly contained **alpha-helices**, which are common in S100 family proteins.
- The structure included **calcium-binding sites (Ca^{2+})** important for protein stability.
- Experimental data and refinement results confirmed **good model quality**.
- The protein appeared **compact, stable, and helical** in 3D view.
- This prediction helped understand the **structure and metal-binding nature** of the protein.



PDB search page showing protein sequence entry.



Structure details of S100P protein (PDB ID: 1J55).

3D ribbon model showing alpha-helical structure.

Data Collection

Overall

ID #	Resolution (High)	Resolution (Low)	Percent Possible (Observed)	R Merge I (Observed)	Net I Over Average Sigma (I)	Redundancy	Number Reflections (All)	Number Reflections (Observed)	Observed Criterion Sigma (F)	Observed Criterion Sigma (I)	B (Isotropic) From Wilson Plot
1	2	40	96.6	0.104	14.9	8.9	57925	57916		-3	11.5

Highest Resolution Shell

ID #	Resolution (High)	Resolution (Low)	Percent Possible (All)	Percent Possible (Observed)	R Merge I (Observed)	Mean I Over Sigma (Observed)	Redundancy	Number Unique Reflections (All)
1	2	2.07	96.6		0.403		8.1	615

Refinement

Statistics

Diffraction ID	Structure Solution Method	Cross Validation method	Resolution (High)	Resolution (Low)	Number Reflections (All)	Number Reflections (Observed)	Number Reflections (R-Free)	Percent Reflections (Observed)	R-Work (Depositor)	R-Free (Depositor)	R-Free Selection Details	Mean Isotropic B
X-RAY DIFFRACTION		THROUGHOUT	2	40	6442	6226	680	96.6	0.214	0.267	random	

RMS Deviations

Key	Refinement Restraint Deviation
c_angle_deg	1.68
c_bond_d	0.019

Non-Hydrogen Atoms Used in Refinement

Non-Hydrogen Atoms	Number
Protein Atoms	673
Nucleic Acid Atoms	
Solvent Atoms	46
Heterogen Atoms	2

Data collection and refinement results.

SOURCE PDB	IDENTIFIER	METHOD X-ray	RESOLUTION	CHAIN	POSITIONS	LINKS
PDB	1J55	X-ray	2.00 Å	A	1-95	PDBe · RCSB-PDB · PDBj · PDBsum · Foldsee
PDB	7NMI	X-ray	2.10 Å	B	1-95	PDBe · RCSB-PDB · PDBj · PDBsum · Foldsee

3D structure databases

AlphaFoldDB | P25815
BMRB | P25815
SMR | P25815
ModBase | Search...
PDBe-KB | Search...

Miscellaneous

EvolutionaryTrace | P25815

X-ray diffraction data for protein.

SOURCE PDB	IDENTIFIER	METHOD NMR	RESOLUTION	CHAIN	POSITIONS	LINKS
PDB	1OZO	NMR		A/B	1-95	PDBe · RCSB-PDB · PDBj · PDBsum · Foldsee
PDB	2MJW	NMR		B/D	1-94	PDBe · RCSB-PDB · PDBj · PDBsum · Foldsee

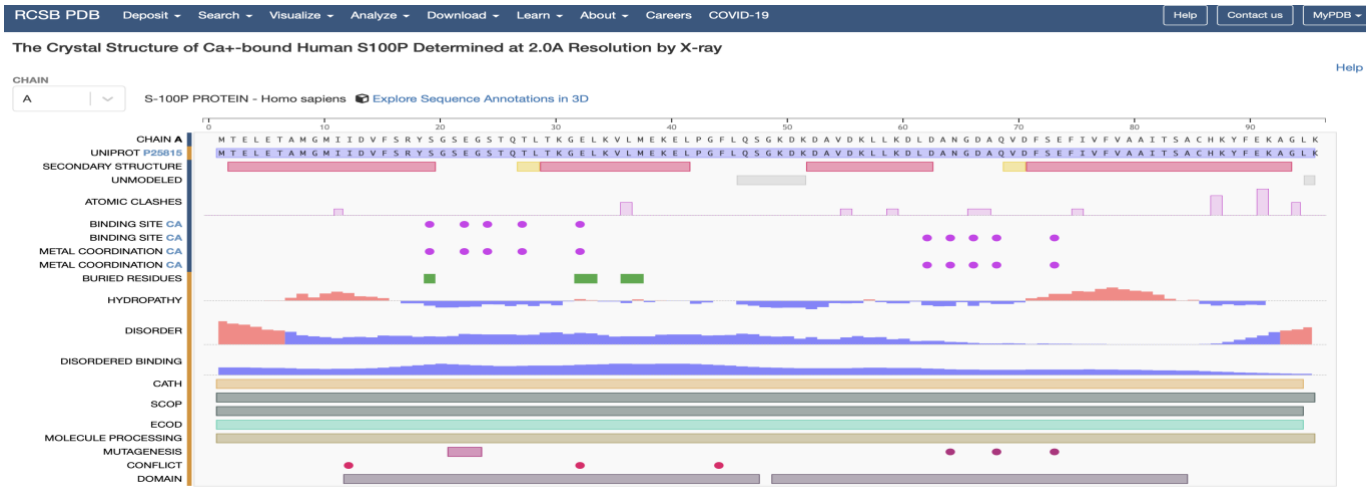
3D structure databases

AlphaFoldDB | P25815
BMRB | P25815
SMR | P25815
ModBase | Search...
PDBe-KB | Search...

Miscellaneous

EvolutionaryTrace | P25815

NMR diffraction data for protein.



Crystal structure by X- ray

The Crystal Structure of Ca+-bound Human S100P Determined at 2.0A Resolution by X-ray

X-RAY DIFFRACTION

Crystallization

Crystalization Experiments

ID	Method	pH	Temperature	Details
1	EVAPORATION	6.5	291	PEG4000, sodium citrate, calcium chloride, pH 6.5, EVAPORATION, temperature 291K

Crystal Properties

Matthews coefficient	Solvent content
2.11	41.83

Crystal Data

Unit Cell

Length (Å)	Angle (°)
a = 60.827	α = 90
b = 60.827	β = 90
c = 47.61	γ = 90

Symmetry

Space Group	P 41 21 2

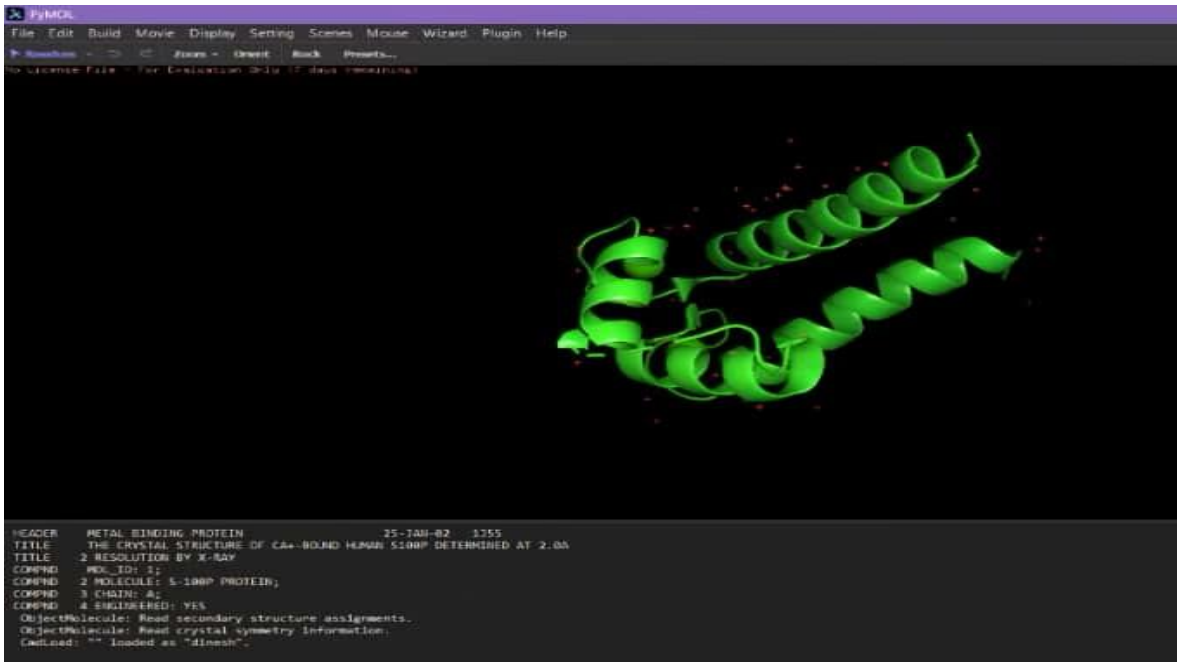
Diffraction

Diffraction Experiment

ID #	Crystal ID	Scattering Type	Data Collection Temperature	Detector	Detector Type	Details	Collection Date	Monochromator	Protocol
1	1	x-ray	110	IMAGE PLATE	MAR scanner 345 mm plate		2001-10-11	M	SINGLE WAVELENGTH

4.5 Visualization Using PyMOL Tool

- The 3D structure of the protein (PDB ID: **1J55**) was opened in **PyMOL** for visualization.
- The protein was shown in **green color** using the **cartoon model**.
- The structure clearly showed **two long alpha-helices** connected by short loops.
- The protein looked **compact and well-folded** in 3D view.
- **Calcium ions (Ca^{2+})** were seen near the helices, showing that it is a **metal-binding protein**.
- The model was rotated and zoomed to view the shape from different angles.

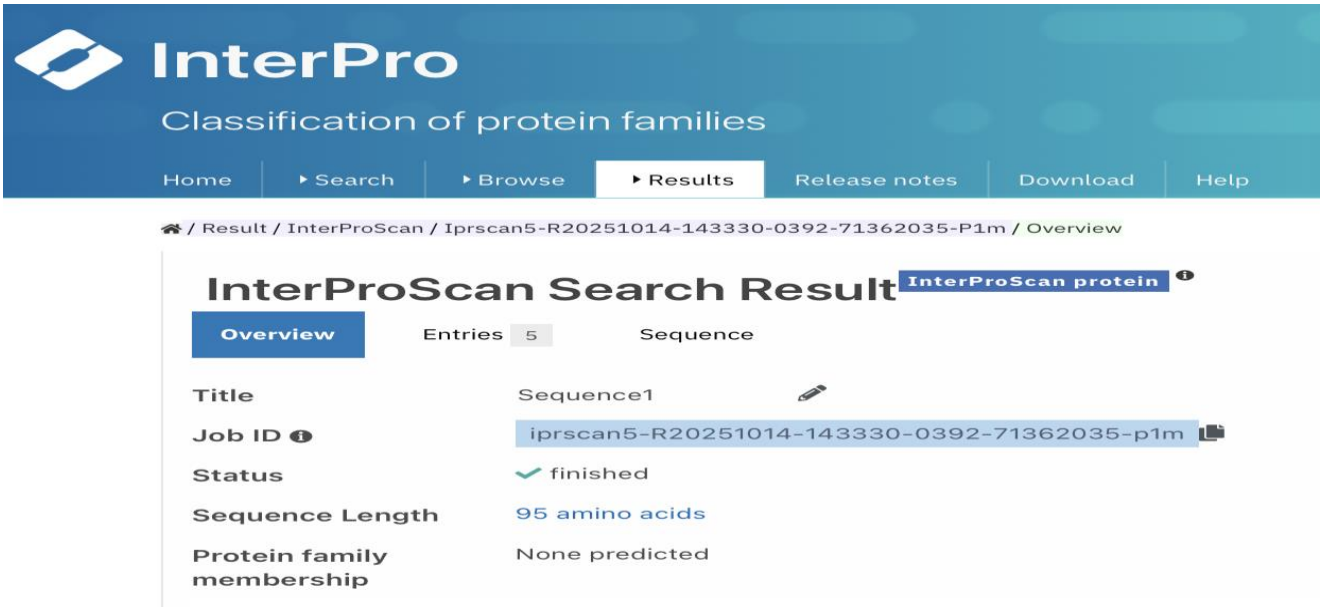


PyMOL visualization of the protein structure (PDB ID: 1J55) showing alpha-helices in cartoon model

InterPro entries showing EF-hand domains and calcium-binding regions.

4.6 Functional and Biological Significance

The **InterProScan** results identified the protein as belonging to the **S100 calcium-binding protein family**, which is known for its role in **calcium ion binding and signal regulation**. The protein contains **EF-hand domains**, which are responsible for binding calcium ions (Ca^{2+}). These domains help the protein to control various cellular processes such as **cell growth, cell movement, and intracellular communication**.



InterProScan overview showing the protein length (95 amino acids) and job completion.

InterProScan Search Result

InterProScan protein

Overview | Entries 5 | Sequence

This protein matches these entries:

1 - 5 of **5** entries matching **InterPro**

Search

Accession	Short Name	Name	Source Database	Matches
IPR001751	S100/CaBP7/8-like_CS	S100/Calcium binding protein 7/8-like, conserved site	InterPro	50
IPR013787	S100_Ca-bd_sub	S100/CaBP-9k-type, calcium binding, subdomain	InterPro	50
IPR034325	S-100_dom	S-100	InterPro	50
IPR011992	EF-hand-dom_pair	EF-hand domain pair	InterPro	50
IPR002048	EF_hand_dom	EF-hand domain	InterPro	50

InterPro GO terms

Biological Process

None

Molecular Function

- transition metal ion binding (GO:0046914)
- calcium ion binding (GO:0005509)

Cellular Component

None

PANTHER GO terms

Biological Process

- endothelial cell migration (GO:0043542)

Molecular Function

- calcium-dependent protein binding (GO:0048306)
- calcium ion binding (GO:0005509)

Cellular Component

- cytoplasm (GO:0005737)
- extracellular exosome (GO:0070062)

GO term results showing calcium ion binding and metal ion binding molecular functions.

ProtParam - Results

REC6_HUMAN (Q69383)

Description:

Endogenous retrovirus group K member 6 Rec protein (Central open reading frame) (c-orf) (cORF) (Endogenous retrovirus K protein 6) (HERV-K(C7) Rec protein) (HERV-K(HML-2.HOM) Rec protein) (HERV-K108 Rec protein) (HERV-K_7p22.1 provirus Rec protein) (K-Rev) (Rev-like protein) (Rev/Rex homolog)

Organism:

Homo sapiens (Human)

The computation has been carried out on the complete sequence (105 amino acids).
Warning: All computation results shown below do **not** take into account any annotated post-translational modification.

[Documentation / Reference]

Number of amino acids: 105
Theoretical pI: 10.22
Molecular weight: 11827.77

Amino acid composition: CSV format

Ala (A)	9	8.6%
Arg (R)	7	6.7%
Asn (N)	6	5.7%
Asp (D)	0	0.0%
Cys (C)	0	0.0%
Gln (Q)	5	4.8%
Glu (E)	9	8.6%
Gly (G)	2	1.9%
His (H)	2	1.9%
Ile (I)	2	1.9%
Leu (L)	9	8.6%
Lys (K)	10	9.5%
Met (M)	8	7.6%
Phe (F)	0	0.0%
Pro (P)	10	9.5%
Ser (S)	8	7.6%
Thr (T)	11	10.5%
Trp (W)	1	1.0%
Tyr (Y)	1	1.0%
Val (V)	5	4.8%
Pyl (O)	0	0.0%
Sec (U)	0	0.0%
(B)	0	0.0%
(Z)	0	0.0%
(X)	0	0.0%

Total number of negatively charged residues (Asp + Glu): 9
Total number of positively charged residues (Arg + Lys): 17

Atomic composition:

Carbon	C	508
Hydrogen	H	854
Nitrogen	N	152
Oxygen	O	155
Sulfur	S	8

Formula: $C_{508}H_{854}N_{152}O_{155}S_{8}$
Total number of atoms: 1677

Extinction coefficients:
Extinction coefficients are in units of M^{-1} cm^{-1}, at 280 nm measured in water.
Ext. coefficient 6990
Abs 0.1% (=1 g/l) 0.591

Estimated half-life:
The N-terminal of the sequence considered is M (Met).
The estimated half-life is: 30 hours (mammalian reticulocytes, in vitro).
>20 hours (yeast, in vivo).
>10 hours (Escherichia coli, in vivo).

Instability index:
The instability index (II) is computed to be 65.65
This classifies the protein as unstable.

Aliphatic index: 63.24
Grand average of hydropathicity (GRAVY):-0.804

ProtParam result showing amino acid composition, molecular weight (11.27 kDa), isoelectric point (pI = 10.2 index (65.65), and GRAVY score (–0.804) for the S100 calcium-binding protein P.

- The protein was studied using the **ProtParam tool**. It has **105 amino acids** and a **molecular weight of 11.27 kDa**. The **isoelectric point (pI)** is **10.22**, which means the protein is **basic** in nature. The protein has more **positively charged amino acids** like lysine and arginine. The **instability index** value (**65.65**) shows that the protein is **slightly unstable**. The **GRAVY score (-0.804)** means it is **hydrophilic** and can dissolve in water. Overall, this protein is **S100 calcium-binding protein P**, which helps in **calcium binding** and **cell signaling**.

5. Results and Discussion

The given protein sequence was studied using different bioinformatics tools.
The **BLASTp** results showed that the protein is **highly similar to other primate proteins**, which means it is **well conserved**.
The **multiple sequence alignment** showed that many amino acids are **same or similar** in all the sequences.
This proves that the protein has an **important function** that is preserved during evolution.

The **phylogenetic tree** showed that primate proteins are grouped closely, showing a **common origin**.
The **3D structure** from **RCSB PDB** and **AlphaFold** showed that the protein mainly has **alpha-helices** and a **compact structure**.
Using **PyMOL**, the 3D model was clearly seen, and **calcium-binding regions** were visible.
The **functional analysis** from **InterPro** and **ProtParam** identified the protein as **S100 calcium-binding protein P**, which helps in **calcium regulation and signaling** in cells.

Overall, the results show that the protein is **structurally stable**, **functionally conserved**, and **biologically important** in primates.
This project also shows how **bioinformatics tools** can be used to study proteins easily and effectively.

6. Conclusion

In this project, different bioinformatics tools were used to study the given protein sequence.
The analysis showed that the protein is **highly conserved among primates** and plays an important role in **calcium regulation**.
The **BLASTp**, **Clustal Omega**, and **Phylogenetic tree** results confirmed the similarity and close relationship between primate proteins.
The **3D structure** from **RCSB PDB** and **AlphaFold** showed that the protein has a **stable alpha-helical structure**.
The **functional analysis** identified it as **S100 calcium-binding protein P**, which is involved in **cell signaling and calcium binding**.
This project helped to understand how **bioinformatics tools** can be used to study the **structure, function, and evolution**of proteins in a simple and effective way.

7. References

1. NCBI BLASTp (Basic Local Alignment Search Tool – Protein)
Used to find homologous protein sequences among primates.

2. Clustal Omega – Multiple Sequence Alignment Tool
Used for aligning protein sequences and identifying conserved regions.

3. MEGA11 (Molecular Evolutionary Genetics Analysis)
Used for viewing and analyzing sequence alignments and evolutionary relationships.

4. Phylogeny.fr / Phylo.io
Used for generating phylogenetic trees from aligned protein sequences.

5. RCSB Protein Data Bank (PDB)
Used for retrieving experimentally determined 3D structures of proteins.

6. AlphaFold Protein Structure Database (EMBL-EBI)
Used for predicted protein 3D structures based on amino acid sequences.

7. PyMOL Molecular Graphics System
Used for visualizing and analyzing protein 3D structures.

8. UniProt Knowledgebase (UniProtKB)
Used for retrieving protein information, function, and biological role.

9. InterPro Database
Used for identifying protein domains and families.

10. ProtParam (ExPASy Server)
Used for calculating physicochemical properties of the protein.